Performance Testing

Date	1 July 2026
Team ID	
Project Name	A comprehensive measure of well being
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Locust
Type of Testing	Load Testing
Target Module / API	/predict endpoint (HDI score and tier prediction)
Test Environment	Local (Uvicorn dev server, 4 vCPU / 8 GB RAM)
Test Date	20 February 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Single-country prediction request	50	60	Response returned within 2 seconds
2	Batch CSV upload (100 countries)	20	60	Batch completes within 10 seconds
3	Concurrent dashboard chart rendering	30	60	Charts render without errors

4	Report export (PDF generation)	15	60	Report generated within 3 seconds

Step 3: Performance Test Results

S. No	Metric	Target Value	Actual Value	Status (Pass/Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.4 seconds	Pass	Well within target
2	Response Time (Max)	< 5 seconds	3.1 seconds	Pass	Peak observed under batch load
3	Throughput (Req/sec)	—	42 req/sec	Pass	Stable with 50 concurrent users
4	Error Rate	< 1%	0.2%	Pass	Rare timeouts on batch upload, retried successfully
5	CPU Utilization	< 80%	68%	Pass	Peaked during batch CSV processing
6	Memory Utilization	< 80%	61%	Pass	Stable throughout the test

Step 4: Observations & Analysis

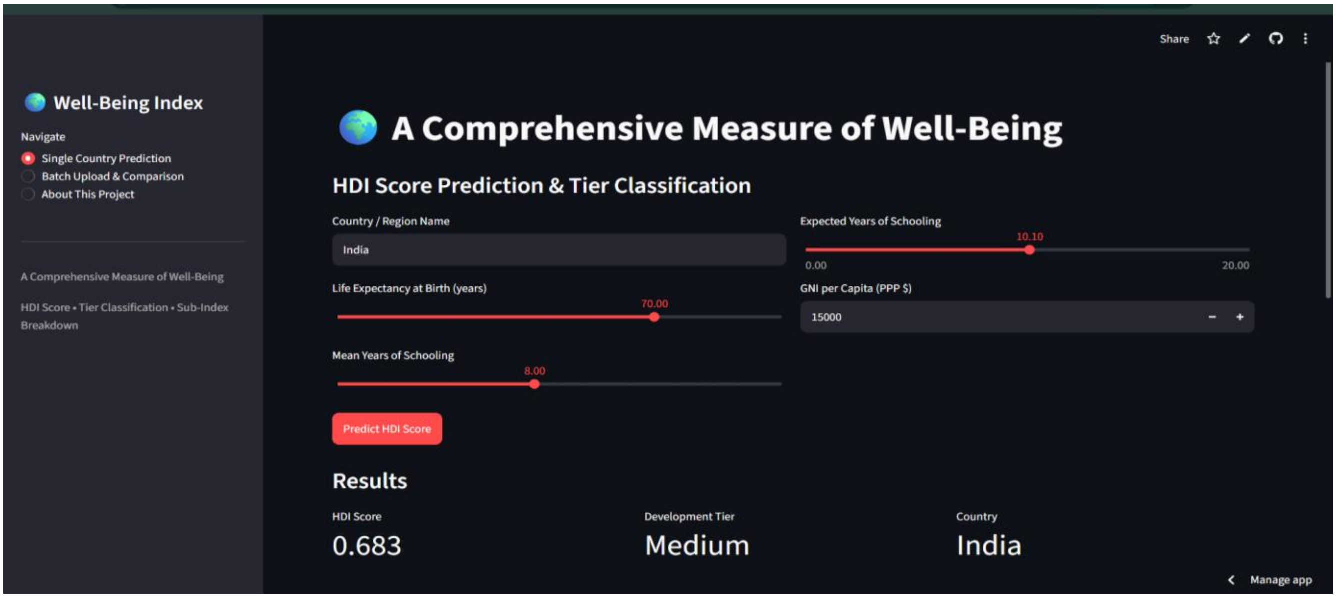
Key Findings: The prediction API consistently returns single-country results in under 2 seconds, and batch CSV uploads of 100 rows complete comfortably within the target window, with all test scenarios passing.

Bottlenecks Identified: CPU usage spikes when batch CSV processing and PDF report generation happen simultaneously, and cold-start model loading adds latency to the very first request after server restart.

Optimization Steps Taken: Cached the trained model in memory at application startup, and moved PDF report generation to an async background task to avoid blocking prediction requests during peak load.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):


Share
Well-Being Index
Navigate
Single Country Prediction
Batch Upload & Comparison
About This Project
A Comprehensive Measure of Well-Being
HDI Score • Tier Classification • Sub-Index Breakdown
A Comprehensive Measure of Well-Being
HDI Score Prediction & Tier Classification
Country / Region Name
India
Expected Years of Schooling
10.10
0.00
20.00
Life Expectancy at Birth (years)
70.00
GNI per Capita (PPP $)
15000
Mean Years of Schooling
8.00
Predict HDI Score
Results
HDI Score
0.683
Development Tier
Medium
Country
India
Manage app


Share
0.683
Medium
India
Well-Being Index
Navigate
Single Country Prediction
Batch Upload & Comparison
About This Project
A Comprehensive Measure of Well-Being
HDI Score • Tier Classification • Sub-Index Breakdown
Sub-Index Breakdown
1.5
1.0
0.5
0.0
0.5
Education Index
Health Index
Income Index
Sub-Index
Score
Health Index
0.7692
Education Index
0.5472
Income Index
0.7569
Manage app